

FagusED · A-Level 9702
1 Physical quantities and units

What is the central idea in Physical quantities and units?

9702-01-C1

QUESTION

FagusED · A-Level 9702
1 Physical quantities and units

Which mechanism, condition or distinction matters in Physical quantities and units?

9702-01-C2

QUESTION

FagusED · A-Level 9702
1 Physical quantities and units

Which quantity has SI base unit kg?
A Mass
B Force
C Pressure
D Energy

9702-01-M

QUESTION

FagusED · A-Level 9702
1 Physical quantities and units

A resistance is calculated from 6.0 ± 0.2 V and 2.0 ± 0.1 A. Determine R and its approximate absolute uncertainty.

9702-01-P

QUESTION

FagusED · A-Level 9702
2 Kinematics

What is the central idea in Kinematics?

9702-02-C1

QUESTION

FagusED · A-Level 9702
2 Kinematics

Which mechanism, condition or distinction matters in Kinematics?

9702-02-C2

QUESTION

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2 Kinematics

What is the gradient of a velocity-time graph?
A Displacement
B Acceleration
C Distance
D Power

9702-02-M

QUESTION

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2 Kinematics

A trolley accelerates uniformly from rest to 9 m s^{-1} over 18 m. Find acceleration and travel time.

9702-02-P

QUESTION

FagusED · A-Level 9702
3 Dynamics

What is the central idea in Dynamics?

9702-03-C1

QUESTION

FagusED · A-Level 9702
3 Dynamics

Which mechanism, condition or distinction matters in Dynamics?

9702-03-C2

QUESTION

FagusED · A-Level 9702
3 Dynamics

For an isolated collision, which total is conserved?
A Kinetic energy always
B Temperature
C Momentum
D Speed of each body

9702-03-M

QUESTION

FagusED · A-Level 9702
3 Dynamics

A 0.6 kg cart at 4 m s^{-1} collides and sticks to a stationary 0.4 kg cart. Find their common velocity and state the condition for conservation.

9702-03-P

QUESTION

FagusED · A-Level 9702
4 Forces, density and pressure

What is the central idea in Forces, density and pressure?

9702-04-C1

QUESTION

FagusED · A-Level 9702
4 Forces, density and pressure

Which mechanism, condition or distinction matters in Forces, density and pressure?

9702-04-C2

QUESTION

FagusED · A-Level 9702
4 Forces, density and pressure

Pressure increase at depth h in a liquid is proportional to what?
A $1/h$
B h^2 only
C Surface area
D pgh

9702-04-M

QUESTION

FagusED · A-Level 9702
1 Physical quantities and units

A. Mass
Why: A physical quantity combines a number and a unit.

9702-01-M

ANSWER

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1 Physical quantities and units

Dimensional homogeneity checks equations, while percentage uncertainties add for products or quotients. Precision describes repeatability; accuracy concerns closeness to the true value.

9702-01-C2

ANSWER

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1 Physical quantities and units

A physical quantity combines a number and a unit. Base SI quantities include mass, length, time and current; derived units are products of base units.

9702-01-C1

ANSWER

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2 Kinematics

The constant-acceleration equations apply only where acceleration remains uniform.

9702-02-C2

ANSWER

FagusED · A-Level 9702
2 Kinematics

Displacement is a vector change in position; velocity is its rate of change. A displacement-time gradient is velocity and velocity-time gradient is acceleration; signed area under velocity-time is displacement.

9702-02-C1

ANSWER

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1 Physical quantities and units

$R = V/I = 3.0 \, \Omega$. Percentage uncertainty = $0.2/6.0 + 0.1/2.0 = 0.0833$. Absolute uncertainty $\approx 0.25 \, \Omega$.

9702-01-P

ANSWER

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3 Dynamics

Newton's second law relates resultant force to the rate of change of momentum: $F = \Delta p/\Delta t$, reducing to ma for constant mass.

9702-03-C1

ANSWER

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2 Kinematics

$v^2 = u^2 + 2as$, giving $a = 81/36 = 2.25 \, \text{m s}^{-2}$; $t = (v - u)/a = 4.0 \, \text{s}$.

9702-02-P

ANSWER

FagusED · A-Level 9702
2 Kinematics

B. Acceleration
Why: Displacement is a vector change in position; velocity is its rate of change.

9702-02-M

ANSWER

FagusED · A-Level 9702
3 Dynamics

Initial momentum $0.6 \times 4 = 2.4 \, \text{kg m s}^{-1}$; common velocity $2.4/(0.6+0.4) = 2.4 \, \text{m s}^{-1}$; negligible external resultant impulse.

9702-03-P

ANSWER

FagusED · A-Level 9702
3 Dynamics

C. Momentum
Why: Newton's second law relates resultant force to the rate of change of momentum: $F = \Delta p/\Delta t$, reducing to ma for constant mass.

9702-03-M

ANSWER

FagusED · A-Level 9702
3 Dynamics

Momentum is a vector and is conserved in an isolated interaction. During a collision, increasing contact time reduces average force for a fixed change in momentum.

9702-03-C2

ANSWER

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4 Forces, density and pressure

D. $pg h$
Why: Force diagrams must show forces on one chosen body.

9702-04-M

ANSWER

FagusED · A-Level 9702
4 Forces, density and pressure

The moment about a pivot is force times perpendicular lever arm. Density is mass/volume and hydrostatic pressure change $\Delta p = \rho g \Delta h$; pressure acts normally to a surface.

9702-04-C2

ANSWER

FagusED · A-Level 9702
4 Forces, density and pressure

Force diagrams must show forces on one chosen body. Resolve vectors before finding the resultant or equilibrium.

9702-04-C1

ANSWER

FagusED · A-Level 9702
4 Forces, density and pressure

A water tank is filled to depth 1.8 m. Find the pressure increase from surface to bottom, using $\rho = 1000 \text{ kg m}^{-3}$ and $g = 9.81 \text{ m s}^{-2}$.

9702-04-P

QUESTION

FagusED · A-Level 9702
5 Work, energy and power

What is the central idea in Work, energy and power?

9702-05-C1

QUESTION

FagusED · A-Level 9702
5 Work, energy and power

Which mechanism, condition or distinction matters in Work, energy and power?

9702-05-C2

QUESTION

FagusED · A-Level 9702
5 Work, energy and power

Which expression gives kinetic energy?

- A $\frac{1}{2}mv^2$
- B mv
- C mgh
- D F/t

9702-05-M

QUESTION

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5 Work, energy and power

A motor lifts a 30 kg mass through 2.0 m in 5.0 s while drawing 150 W. Calculate efficiency.

9702-05-P

QUESTION

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6 Deformation of solids

What is the central idea in Deformation of solids?

9702-06-C1

QUESTION

FagusED · A-Level 9702
6 Deformation of solids

Which mechanism, condition or distinction matters in Deformation of solids?

9702-06-C2

QUESTION

FagusED · A-Level 9702
6 Deformation of solids

Which ratio gives Young modulus?

- A Strain/stress
- B Stress/strain
- C Force/extension only
- D Extension/length only

9702-06-M

QUESTION

FagusED · A-Level 9702
6 Deformation of solids

A spring has $k = 80 \text{ N m}^{-1}$ and extends by 0.050 m. Find force and elastic energy stored.

9702-06-P

QUESTION

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7 Waves

What is the central idea in Waves?

9702-07-C1

QUESTION

FagusED · A-Level 9702
7 Waves

Which mechanism, condition or distinction matters in Waves?

9702-07-C2

QUESTION

FagusED · A-Level 9702
7 Waves

How is wave speed related to frequency and wavelength?

- A $v=f/\lambda$
- B $v=\lambda/f$
- C $v=f\lambda$
- D $v=f+\lambda$

9702-07-M

QUESTION

FagusED · A-Level 9702
7 Waves

A wave travels at 340 m s^{-1} and has frequency 680 Hz. Find wavelength and period.

9702-07-P

QUESTION

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8 Superposition

What is the central idea in Superposition?

9702-08-C1

QUESTION

FagusED · A-Level 9702
8 Superposition

Which mechanism, condition or distinction matters in Superposition?

9702-08-C2

QUESTION

FagusED · A-Level 9702
5 Work, energy and power

Power measures energy transferred per unit time. For a real process, conservation applies to all stores even when useful efficiency is below 100%.

9702-05-C2

ANSWER

FagusED · A-Level 9702
5 Work, energy and power

Work is energy transfer by a force through displacement in its direction: $W = Fs \cos \theta$. Kinetic energy is $\frac{1}{2}mv^2$; gravitational energy change near Earth is $mg\Delta h$.

9702-05-C1

ANSWER

FagusED · A-Level 9702
4 Forces, density and pressure

$$\Delta p = \rho gh = 1000 \times 9.81 \times 1.8 = 1.77 \times 10^4 \text{ Pa (3 s.f.)}$$

9702-04-P

ANSWER

FagusED · A-Level 9702
6 Deformation of solids

Stress = force/cross-sectional area; strain = extension/original length. Young modulus E = stress/strain in a linear elastic region.

9702-06-C1

ANSWER

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5 Work, energy and power

$$\text{Useful power} = mgh/t = 30 \times 9.81 \times 2/5 = 117.7 \text{ W. Efficiency} = 117.7/150 = 0.785, \text{ or } 78.5\%.$$

9702-05-P

ANSWER

FagusED · A-Level 9702
5 Work, energy and power

A. $\frac{1}{2}mv^2$
Why: Work is energy transfer by a force through displacement in its direction: $W = Fs \cos \theta$.

9702-05-M

ANSWER

FagusED · A-Level 9702
6 Deformation of solids

$$F = kx = 4.0 \text{ N. Energy} = \frac{1}{2}kx^2 = 0.10 \text{ J.}$$

9702-06-P

ANSWER

FagusED · A-Level 9702
6 Deformation of solids

B. Stress/strain
Why: Stress = force/cross-sectional area; strain = extension/original length.

9702-06-M

ANSWER

FagusED · A-Level 9702
6 Deformation of solids

The gradient of a force-extension graph is spring constant; its area gives work done. Elastic deformation is reversible on unloading, whereas plastic deformation leaves a permanent extension.

9702-06-C2

ANSWER

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7 Waves

C. $v = f\lambda$
Why: A progressive wave transfers energy without a net transport of matter.

9702-07-M

ANSWER

FagusED · A-Level 9702
7 Waves

Intensity scales as amplitude squared for otherwise comparable waves. Superposition adds simultaneous displacements algebraically.

9702-07-C2

ANSWER

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7 Waves

A progressive wave transfers energy without a net transport of matter. Its speed is $v = f\lambda$. Transverse displacement is perpendicular to propagation; longitudinal displacement is parallel.

9702-07-C1

ANSWER

FagusED · A-Level 9702
8 Superposition

Diffraction spreads waves after an aperture, especially where width is comparable with wavelength. Stationary waves form nodes and antinodes from counter-propagating waves.

9702-08-C2

ANSWER

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8 Superposition

Interference needs coherent sources with a stable phase difference. Constructive interference arises at path differences $n\lambda$ and destructive at $(n + \frac{1}{2})\lambda$ for in-phase sources.

9702-08-C1

ANSWER

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7 Waves

$$\lambda = v/f = 0.50 \text{ m; } T = 1/f = 1.47 \times 10^{-3} \text{ s.}$$

9702-07-P

ANSWER

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8 Superposition

For coherent in-phase waves, which path difference yields a bright fringe?

- A $\lambda/2$
- B $3\lambda/2$
- C $\lambda/4$
- D λ

9702-08-M

QUESTION

FagusED · A-Level 9702
8 Superposition

Two slits are 0.30 mm apart and 1.5 m from a screen. Adjacent bright fringes are 3.0 mm apart. Find wavelength.

9702-08-P

QUESTION

FagusED · A-Level 9702
9 Electricity

What is the central idea in Electricity?

9702-09-C1

QUESTION

FagusED · A-Level 9702
9 Electricity

Which mechanism, condition or distinction matters in Electricity?

9702-09-C2

QUESTION

FagusED · A-Level 9702
9 Electricity

For a resistor at fixed conditions, which expression gives power?

- A I^2R
- B I/R
- C R/V
- D Q/t^2

9702-09-M

QUESTION

FagusED · A-Level 9702
9 Electricity

A wire of length 4.0 m and area $2.0 \times 10^{-6} \text{ m}^2$ has resistance 0.10 Ω . Determine resistivity.

9702-09-P

QUESTION

FagusED · A-Level 9702
10 D.C. circuits

What is the central idea in D.C. circuits?

9702-10-C1

QUESTION

FagusED · A-Level 9702
10 D.C. circuits

Which mechanism, condition or distinction matters in D.C. circuits?

9702-10-C2

QUESTION

FagusED · A-Level 9702
10 D.C. circuits

How is terminal p.d. of a discharging cell related to e.m.f.?

- A $V = \epsilon + Ir$
- B $V = \epsilon - Ir$
- C $V = Ir - \epsilon$
- D $V = \epsilon/I$

9702-10-M

QUESTION

FagusED · A-Level 9702
10 D.C. circuits

A cell has $\epsilon = 9.0 \text{ V}$ and internal resistance 0.50 Ω ; external resistance is 4.0 Ω . Find current and terminal p.d.

9702-10-P

QUESTION

FagusED · A-Level 9702
11 Particle physics

What is the central idea in Particle physics?

9702-11-C1

QUESTION

FagusED · A-Level 9702
11 Particle physics

Which mechanism, condition or distinction matters in Particle physics?

9702-11-C2

QUESTION

FagusED · A-Level 9702
11 Particle physics

What is the quark content of a proton?

- A udd
- B uuu
- C uud
- D ddd

9702-11-M

QUESTION

FagusED · A-Level 9702
11 Particle physics

In β^- decay, a neutron becomes a proton, electron and electron antineutrino. Verify charge and lepton-number conservation.

9702-11-P

QUESTION

FagusED · A-Level 9702
12 Motion in a circle

What is the central idea in Motion in a circle?

9702-12-C1

QUESTION

Electric current is charge flow rate
 $I = \Delta Q / \Delta t$. Potential difference is energy transferred per unit charge, $V = W/Q$.

9702-09-C1

ANSWER

Use $x = \lambda D/a$, so $\lambda = xa/D = (3.0 \times 10^{-3})(3.0 \times 10^{-4})/1.5 = 6.0 \times 10^{-7} \text{ m}$.

9702-08-P

ANSWER

D. λ
 Why: Interference needs coherent sources with a stable phase difference.

9702-08-M

ANSWER

$\rho = RA/L = 0.10 \times 2.0 \times 10^{-6}/4.0 = 5.0 \times 10^{-8} \Omega \text{ m}$.

9702-09-P

ANSWER

A. I^2R
 Why: Electric current is charge flow rate $I = \Delta Q / \Delta t$.

9702-09-M

ANSWER

Resistance $R = V/I$; an ohmic conductor at constant temperature has a linear I-V graph through the origin. Electrical power $P = VI = I^2R$. Resistivity depends on material and temperature.

9702-09-C2

ANSWER

B. $V = \epsilon - Ir$
 Why: Kirchhoff's first law conserves charge at a junction; the second conserves energy around a loop.

9702-10-M

ANSWER

In series, current is shared and potential differences add; in parallel, potential difference is shared and currents add. A source with internal resistance r has terminal p.d. $V = \epsilon - Ir$ during discharge.

9702-10-C2

ANSWER

Kirchhoff's first law conserves charge at a junction; the second conserves energy around a loop.

9702-10-C1

ANSWER

Particle reactions must conserve charge, baryon number and lepton number; mass-energy may be converted according to $E = mc^2$.

9702-11-C2

ANSWER

Hadrons are made of quarks: baryons contain three quarks and mesons a quark-antiquark pair. Leptons, including electrons and neutrinos, are fundamental.

9702-11-C1

ANSWER

$I = \epsilon / (R + r) = 9/4.5 = 2.0 \text{ A}$; $V = IR = 8.0 \text{ V}$.

9702-10-P

ANSWER

Circular motion changes velocity direction even at constant speed. The centripetal acceleration $v^2/r = \omega^2 r$ points toward the centre, and its resultant force is mv^2/r .

9702-12-C1

ANSWER

Charge: $0 \rightarrow +1 -1 +0 = 0$.
 Electron lepton number: $0 \rightarrow +1 -1 = 0$. Baryon number remains +1.

9702-11-P

ANSWER

C. uud
 Why: Hadrons are made of quarks: baryons contain three quarks and mesons a quark-antiquark pair.

9702-11-M

ANSWER

FagusED · A-Level 9702
12 Motion in a circle

Which mechanism, condition or distinction matters in Motion in a circle?

9702-12-C2

QUESTION

FagusED · A-Level 9702
12 Motion in a circle

What direction is centripetal acceleration?
A Tangential to motion
B Away from centre
C Vertical always
D Towards centre

9702-12-M

QUESTION

FagusED · A-Level 9702
12 Motion in a circle

An object travels a horizontal circle of radius 0.80 m in 2.0 s. Determine angular speed and centripetal acceleration.

9702-12-P

QUESTION

FagusED · A-Level 9702
13 Gravitational fields

What is the central idea in Gravitational fields?

9702-13-C1

QUESTION

FagusED · A-Level 9702
13 Gravitational fields

Which mechanism, condition or distinction matters in Gravitational fields?

9702-13-C2

QUESTION

FagusED · A-Level 9702
13 Gravitational fields

How does gravitational field strength vary outside a spherical mass?
A As $1/r^2$
B As r^2
C As $1/r$
D Constant with r

9702-13-M

QUESTION

FagusED · A-Level 9702
13 Gravitational fields

At radius r , orbital speed is v . Derive the new speed for a circular orbit of radius $4r$ about the same central mass.

9702-13-P

QUESTION

FagusED · A-Level 9702
14 Temperature

What is the central idea in Temperature?

9702-14-C1

QUESTION

FagusED · A-Level 9702
14 Temperature

Which mechanism, condition or distinction matters in Temperature?

9702-14-C2

QUESTION

FagusED · A-Level 9702
14 Temperature

Which conversion to kelvin is closest for $27\text{ }^{\circ}\text{C}$?
A 27 K
B 300 K
C 246 K
D 373 K

9702-14-M

QUESTION

FagusED · A-Level 9702
14 Temperature

A gas warms from $20\text{ }^{\circ}\text{C}$ to $80\text{ }^{\circ}\text{C}$. Give initial and final absolute temperatures and temperature rise in kelvin.

9702-14-P

QUESTION

FagusED · A-Level 9702
15 Ideal gases

What is the central idea in Ideal gases?

9702-15-C1

QUESTION

FagusED · A-Level 9702
15 Ideal gases

Which mechanism, condition or distinction matters in Ideal gases?

9702-15-C2

QUESTION

FagusED · A-Level 9702
15 Ideal gases

Which is the ideal-gas equation?
A $p/T=nRV$
B $pV=nR/T$
C $pV=nRT$
D $p=nVRT$

9702-15-M

QUESTION

FagusED · A-Level 9702
15 Ideal gases

A sealed rigid vessel at 300 K has $p = 100\text{ kPa}$. Find p at 450 K and state the assumption.

9702-15-P

QUESTION

FagusED · A-Level 9702
12 Motion in a circle

$\omega = 2\pi/T = \pi \text{ rad s}^{-1}$; $a = \omega^2 r = 7.90 \text{ m s}^{-2}$.

9702-12-P

ANSWER

FagusED · A-Level 9702
12 Motion in a circle

D. Towards centre
Why: Circular motion changes velocity direction even at constant speed.

9702-12-M

ANSWER

FagusED · A-Level 9702
12 Motion in a circle

Angular speed $\omega = 2\pi/T$.
“Centripetal force” describes a resultant supplied by real forces, not an extra interaction.

9702-12-C2

ANSWER

FagusED · A-Level 9702
13 Gravitational fields

A. As $1/r^2$
Why: Gravitational field strength is force per unit mass, $g = GM/r^2$ outside a spherical mass.

9702-13-M

ANSWER

FagusED · A-Level 9702
13 Gravitational fields

The negative sign records an attractive bound state. Circular satellites satisfy $GMm/r^2 = mv^2/r$; a geostationary orbit has Earth's rotational period.

9702-13-C2

ANSWER

FagusED · A-Level 9702
13 Gravitational fields

Gravitational field strength is force per unit mass, $g = GM/r^2$ outside a spherical mass. Potential is work per unit mass relative to infinity, $\phi = -GM/r$.

9702-13-C1

ANSWER

FagusED · A-Level 9702
14 Temperature

A thermometric property must vary predictably with temperature; a calibrated scale turns property readings into temperatures. ΔT in kelvin equals ΔT in $^{\circ}\text{C}$.

9702-14-C2

ANSWER

FagusED · A-Level 9702
14 Temperature

Temperature measures thermal state and determines direction of net thermal energy transfer. At absolute zero kelvin is zero, though particles still have quantum behaviour.

9702-14-C1

ANSWER

FagusED · A-Level 9702
13 Gravitational fields

Equate $GMm/r^2 = mv^2/r$ to give $v = \sqrt{GM/r}$. At $4r$, speed is $v/2$.

9702-13-P

ANSWER

FagusED · A-Level 9702
15 Ideal gases

The ideal gas equation $pV = nRT$ links macroscopic state variables. In kinetic theory, pressure arises from collisions and mean translational kinetic energy per molecule is $3kT/2$.

9702-15-C1

ANSWER

FagusED · A-Level 9702
14 Temperature

Initial = 293 K; final = 353 K; rise = 60 K.

9702-14-P

ANSWER

FagusED · A-Level 9702
14 Temperature

B. 300 K
Why: Temperature measures thermal state and determines direction of net thermal energy transfer.

9702-14-M

ANSWER

FagusED · A-Level 9702
15 Ideal gases

At fixed volume and amount, p/T is constant. $p_2 = 100 \times 450/300 = 150 \text{ kPa}$. Assume ideal gas behaviour.

9702-15-P

ANSWER

FagusED · A-Level 9702
15 Ideal gases

C. $pV = nRT$
Why: The ideal gas equation $pV = nRT$ links macroscopic state variables.

9702-15-M

ANSWER

FagusED · A-Level 9702
15 Ideal gases

Use kelvin, pascals and cubic metres; a real gas approaches ideal behaviour at low pressure and high temperature.

9702-15-C2

ANSWER

FagusED · A-Level 9702
16 Thermodynamics

What is the central idea in Thermodynamics?

9702-16-C1

QUESTION

FagusED · A-Level 9702
16 Thermodynamics

Which mechanism, condition or distinction matters in Thermodynamics?

9702-16-C2

QUESTION

FagusED · A-Level 9702
16 Thermodynamics

A gas receives heat Q and does work W_{out} . What is ΔU ?
A $Q + W_{\text{out}}$
B $W_{\text{out}} - Q$
C $-Q - W_{\text{out}}$
D $Q - W_{\text{out}}$

9702-16-M

QUESTION

FagusED · A-Level 9702
16 Thermodynamics

A gas receives 400 J of heating and does 150 J of work on its surroundings. Find ΔU and explain the sign of the work term.

9702-16-P

QUESTION

FagusED · A-Level 9702
17 Oscillations

What is the central idea in Oscillations?

9702-17-C1

QUESTION

FagusED · A-Level 9702
17 Oscillations

Which mechanism, condition or distinction matters in Oscillations?

9702-17-C2

QUESTION

FagusED · A-Level 9702
17 Oscillations

Which expression describes SHM acceleration?
A $a = -\omega^2 x$
B $a = +\omega^2 x$
C $a = \omega x^2$
D $a = 0$ throughout

9702-17-M

QUESTION

FagusED · A-Level 9702
17 Oscillations

A spring oscillator has mass 0.50 kg and $k = 200 \text{ N m}^{-1}$. Find period and describe the effect of quadrupling mass.

9702-17-P

QUESTION

FagusED · A-Level 9702
18 Electric fields

What is the central idea in Electric fields?

9702-18-C1

QUESTION

FagusED · A-Level 9702
18 Electric fields

Which mechanism, condition or distinction matters in Electric fields?

9702-18-C2

QUESTION

FagusED · A-Level 9702
18 Electric fields

Between uniform parallel plates, E equals what?
A Vd
B V/d
C d/V
D V^2d

9702-18-M

QUESTION

FagusED · A-Level 9702
18 Electric fields

Parallel plates 5.0 mm apart have p.d. 300 V. Find field strength and work done moving an electron across the p.d.

9702-18-P

QUESTION

FagusED · A-Level 9702
19 Capacitance

What is the central idea in Capacitance?

9702-19-C1

QUESTION

FagusED · A-Level 9702
19 Capacitance

Which mechanism, condition or distinction matters in Capacitance?

9702-19-C2

QUESTION

FagusED · A-Level 9702
19 Capacitance

For capacitance C at p.d. V , energy stored is what?
A CV^2
B $2CV^2$
C $\frac{1}{2}CV^2$
D $\frac{1}{2}C/V^2$

9702-19-M

QUESTION

FagusED · A-Level 9702
16 Thermodynamics

D. $Q - W_{\text{out}}$

Why: Internal energy is the total random microscopic kinetic and potential energy of particles.

9702-16-M

ANSWER

FagusED · A-Level 9702
16 Thermodynamics

During melting at constant temperature, energy increases molecular potential energy. Specific heat capacity relates heating to $mc\Delta T$; latent heat relates phase change to mL .

9702-16-C2

ANSWER

FagusED · A-Level 9702
16 Thermodynamics

Internal energy is the total random microscopic kinetic and potential energy of particles. First law: $\Delta U = Q + W$ when W is work done on the system (check sign convention).

9702-16-C1

ANSWER

FagusED · A-Level 9702
17 Oscillations

At equilibrium speed is maximum; at extreme displacement speed is zero. Resonance occurs when driving frequency is close to natural frequency.

9702-17-C2

ANSWER

FagusED · A-Level 9702
17 Oscillations

Simple harmonic motion has acceleration proportional to displacement and opposite in direction: $a = -\omega^2 x$. A mass-spring system has $T = 2\pi\sqrt{m/k}$ when damping is negligible.

9702-17-C1

ANSWER

FagusED · A-Level 9702
16 Thermodynamics

Work done on gas = -150 J . $\Delta U = Q + W = 400 - 150 = +250 \text{ J}$; expansion transfers energy out of the gas.

9702-16-P

ANSWER

FagusED · A-Level 9702
18 Electric fields

Electric field strength E is force per unit positive charge. Between parallel plates, uniform $E = V/d$; a point charge produces a radial field proportional to $1/r^2$.

9702-18-C1

ANSWER

FagusED · A-Level 9702
17 Oscillations

$T = 2\pi\sqrt{m/k} = 2\pi\sqrt{(0.50/200)} \approx 0.314 \text{ s}$. Quadrupling m doubles T .

9702-17-P

ANSWER

FagusED · A-Level 9702
17 Oscillations

A. $a = -\omega^2 x$
Why: Simple harmonic motion has acceleration proportional to displacement and opposite in direction: $a = -\omega^2 x$.

9702-17-M

ANSWER

FagusED · A-Level 9702
18 Electric fields

$E = V/d = 300/0.005 = 6.0 \times 10^4 \text{ V m}^{-1}$; energy magnitude $eV = 1.60 \times 10^{-19} \times 300 = 4.8 \times 10^{-17} \text{ J}$.

9702-18-P

ANSWER

FagusED · A-Level 9702
18 Electric fields

B. V/d
Why: Electric field strength E is force per unit positive charge.

9702-18-M

ANSWER

FagusED · A-Level 9702
18 Electric fields

Electric potential is energy per unit charge. A positive charge tends to move in the direction of decreasing potential when free.

9702-18-C2

ANSWER

FagusED · A-Level 9702
19 Capacitance

C. $\frac{1}{2}CV^2$
Why: Capacitance $C = Q/V$ measures charge stored per potential difference.

9702-19-M

ANSWER

FagusED · A-Level 9702
19 Capacitance

During resistor-capacitor discharge, $V = V_0 e^{(-t/RC)}$; the time constant RC is the time for voltage to fall to $1/e$ of its initial value.

9702-19-C2

ANSWER

FagusED · A-Level 9702
19 Capacitance

Capacitance $C = Q/V$ measures charge stored per potential difference. A capacitor stores energy $E = \frac{1}{2}QV = \frac{1}{2}CV^2$.

9702-19-C1

ANSWER

FagusED · A-Level 9702
19 Capacitance

A 220 μF capacitor discharges through 10 k Ω . Find its time constant and fraction of voltage remaining after one time constant.

9702-19-P

QUESTION

FagusED · A-Level 9702
20 Magnetic fields

What is the central idea in Magnetic fields?

9702-20-C1

QUESTION

FagusED · A-Level 9702
20 Magnetic fields

Which mechanism, condition or distinction matters in Magnetic fields?

9702-20-C2

QUESTION

FagusED · A-Level 9702
20 Magnetic fields

Magnetic force on a moving charge at right angles to B is what?

- A Bq/v
- B Bv/q
- C qv/B
- D Bqv

9702-20-M

QUESTION

FagusED · A-Level 9702
20 Magnetic fields

A 50-turn coil's flux per turn changes from 0.020 to 0.0050 Wb in 0.10 s. Find mean induced e.m.f. magnitude.

9702-20-P

QUESTION

FagusED · A-Level 9702
21 Alternating currents

What is the central idea in Alternating currents?

9702-21-C1

QUESTION

FagusED · A-Level 9702
21 Alternating currents

Which mechanism, condition or distinction matters in Alternating currents?

9702-21-C2

QUESTION

FagusED · A-Level 9702
21 Alternating currents

Which is RMS voltage of a sinusoid with peak voltage V_0 ?

- A $V_0/\sqrt{2}$
- B $\sqrt{2}V_0$
- C $V_0/2$
- D $2V_0$

9702-21-M

QUESTION

FagusED · A-Level 9702
21 Alternating currents

An ideal transformer has 1000 primary turns, 50 secondary turns and 240 V primary RMS. Determine secondary voltage and primary current if secondary current is 2.0 A.

9702-21-P

QUESTION

FagusED · A-Level 9702
22 Quantum physics

What is the central idea in Quantum physics?

9702-22-C1

QUESTION

FagusED · A-Level 9702
22 Quantum physics

Which mechanism, condition or distinction matters in Quantum physics?

9702-22-C2

QUESTION

FagusED · A-Level 9702
22 Quantum physics

What increases the maximum kinetic energy of photoelectrons above threshold?

- A Light intensity alone
- B Light frequency
- C Illuminated area alone
- D Exposure time alone

9702-22-M

QUESTION

FagusED · A-Level 9702
22 Quantum physics

A metal's threshold frequency is 5.0×10^{14} Hz. Light of 8.0×10^{14} Hz falls on it. Find maximum kinetic energy, $h = 6.63 \times 10^{-34}$ J s.

9702-22-P

QUESTION

FagusED · A-Level 9702
23 Nuclear physics

What is the central idea in Nuclear physics?

9702-23-C1

QUESTION

FagusED · A-Level 9702
23 Nuclear physics

Which mechanism, condition or distinction matters in Nuclear physics?

9702-23-C2

QUESTION

FagusED · A-Level 9702
20 Magnetic fields

Faraday's law relates induced e.m.f. to the rate of change of magnetic flux linkage, with Lenz's law setting the opposing direction.

9702-20-C2

ANSWER

FagusED · A-Level 9702
20 Magnetic fields

A current-carrying conductor in a magnetic field experiences $F = BIL \sin \theta$. A moving charge experiences $F = Bqv \sin \theta$; the force is perpendicular to velocity and does no work.

9702-20-C1

ANSWER

FagusED · A-Level 9702
19 Capacitance

$\tau = RC = 10\,000 \times 220 \times 10^{-6} = 2.2 \text{ s}$; fraction $e^{-1} \approx 0.368$.

9702-19-P

ANSWER

FagusED · A-Level 9702
21 Alternating currents

An alternating voltage changes sign periodically. For a sinusoid $V_{\text{rms}} = V_{\text{peak}}/\sqrt{2}$; resistive power uses RMS current and voltage.

9702-21-C1

ANSWER

FagusED · A-Level 9702
20 Magnetic fields

$\Delta(N\Phi) = 50(0.020 - 0.0050) = 0.75 \text{ Wb-turn}$; $\text{abs}(\epsilon) = \Delta(N\Phi)/\Delta t = 7.5 \text{ V}$.

9702-20-P

ANSWER

FagusED · A-Level 9702
20 Magnetic fields

D. Bqv
Why: A current-carrying conductor in a magnetic field experiences $F = BIL \sin \theta$.

9702-20-M

ANSWER

FagusED · A-Level 9702
21 Alternating currents

$V_s = 240 \times 50/1000 = 12 \text{ V}$;
equal powers: $240 I_p = 12 \times 2.0$,
 $I_p = 0.10 \text{ A}$.

9702-21-P

ANSWER

FagusED · A-Level 9702
21 Alternating currents

A. $V_0/\sqrt{2}$
Why: An alternating voltage changes sign periodically.

9702-21-M

ANSWER

FagusED · A-Level 9702
21 Alternating currents

An ideal transformer obeys $V_s/V_p = N_s/N_p$ and conserves input/output power. Transmission at high voltage lowers current and I^2R losses for given power.

9702-21-C2

ANSWER

FagusED · A-Level 9702
22 Quantum physics

B. Light frequency
Why: A photon has energy hf and momentum h/λ .

9702-22-M

ANSWER

FagusED · A-Level 9702
22 Quantum physics

Raising intensity increases emitted electron count at fixed frequency above threshold, while raising frequency increases maximum kinetic energy.

9702-22-C2

ANSWER

FagusED · A-Level 9702
22 Quantum physics

A photon has energy hf and momentum h/λ . Photoelectric emission requires photon frequency above a material's threshold: maximum kinetic energy = $hf - \text{work function}$.

9702-22-C1

ANSWER

FagusED · A-Level 9702
23 Nuclear physics

Half-life is the time for half the undecayed nuclei to remain; decay is random for one nucleus but predictable statistically for a large sample.

9702-23-C2

ANSWER

FagusED · A-Level 9702
23 Nuclear physics

Nuclear mass defect is the difference between separate constituent masses and nuclear mass; binding energy is Δmc^2 . Activity is decay events per second, $A = \lambda N$.

9702-23-C1

ANSWER

FagusED · A-Level 9702
22 Quantum physics

Use $K_{\text{max}} = h(f - f_0)$; frequency difference $3.0 \times 10^{14} \text{ Hz}$; $K_{\text{max}} = 1.99 \times 10^{-19} \text{ J}$.

9702-22-P

ANSWER

FagusED · A-Level 9702
23 Nuclear physics

What is one becquerel?
A One joule per second
B One nucleus per kilogram
C One decay per second
D One half-life per second

9702-23-M

QUESTION

FagusED · A-Level 9702
23 Nuclear physics

A sample starts at 640 Bq and its half-life is 5.0 h. Find activity after 15 h, then state what one becquerel means.

9702-23-P

QUESTION

FagusED · A-Level 9702
24 Medical physics

What is the central idea in Medical physics?

9702-24-C1

QUESTION

FagusED · A-Level 9702
24 Medical physics

Which mechanism, condition or distinction matters in Medical physics?

9702-24-C2

QUESTION

FagusED · A-Level 9702
24 Medical physics

Why is ultrasound echo time halved to calculate depth?
A The probe halves frequency
B The echo is absorbed twice
C Speed halves in tissue
D The sound makes a return trip

9702-24-M

QUESTION

FagusED · A-Level 9702
24 Medical physics

An echo returns 120 μs after emission; sound speed in tissue is 1540 m s^{-1} . Find depth and explain factor 2.

9702-24-P

QUESTION

FagusED · A-Level 9702
25 Astronomy and cosmology

What is the central idea in Astronomy and cosmology?

9702-25-C1

QUESTION

FagusED · A-Level 9702
25 Astronomy and cosmology

Which mechanism, condition or distinction matters in Astronomy and cosmology?

9702-25-C2

QUESTION

FagusED · A-Level 9702
25 Astronomy and cosmology

If distance to a star doubles, its flux becomes what fraction?
A One quarter
B One half
C Two times
D Four times

9702-25-M

QUESTION

FagusED · A-Level 9702
25 Astronomy and cosmology

A stellar spectrum peaks at 500 nm. Estimate surface temperature with Wien constant $2.90 \times 10^{-3} \text{ m K}$.

9702-25-P

QUESTION

FagusED · A-Level 9702
P3 Advanced practical skills

What is the central idea in P3 Advanced practical skills?

9702-26-C1

QUESTION

FagusED · A-Level 9702
P3 Advanced practical skills

Which mechanism, condition or distinction matters in P3 Advanced practical skills?

9702-26-C2

QUESTION

FagusED · A-Level 9702
P3 Advanced practical skills

In a fair pendulum trial, which variable should remain controlled?
A String length if studying length
B Room temperature only
C Number of pages in notes
D Date of experiment

9702-26-M

QUESTION

FagusED · A-Level 9702
P3 Advanced practical skills

A pendulum experiment records times for 10 oscillations. Explain why timing several oscillations improves the estimate of period, and name two controlled variables.

9702-26-P

QUESTION

FagusED · A-Level 9702
P5 Planning, analysis and evaluation

What is the central idea in P5 Planning, analysis and evaluation?

9702-27-C1

QUESTION

Ultrasound echoes locate boundaries because acoustic impedance mismatch reflects part of a pulse; depth equals speed times echo time divided by two. X-ray attenuation depends on material and thickness.

9702-24-C1

ANSWER

15 h is three half-lives. $A = 640/2^3 = 80$ Bq. One Bq is one nuclear decay per second.

9702-23-P

ANSWER

C. One decay per second
Why: Nuclear mass defect is the difference between separate constituent masses and nuclear mass; binding energy is Δmc^2 .

9702-23-M

ANSWER

$d = vt/2 = 1540 \times 120 \times 10^{-6}/2 = 0.0924$ m; measured time includes outward and return travel.

9702-24-P

ANSWER

D. The sound makes a return trip
Why: Ultrasound echoes locate boundaries because acoustic impedance mismatch reflects part of a pulse; depth equals speed times echo time divided by two.

9702-24-M

ANSWER

CT reconstructs cross-sections from many projections; PET detects pairs of annihilation photons to map tracer distribution. Doses require benefit-risk assessment.

9702-24-C2

ANSWER

A. One quarter
Why: Stellar luminosity is emitted power; observed flux falls as $L/(4\pi d^2)$ for isotropic emission.

9702-25-M

ANSWER

Cosmological redshift is evidence of expansion; for relatively nearby galaxies, $v \approx H_0 d$, with limitations at very large distances.

9702-25-C2

ANSWER

Stellar luminosity is emitted power; observed flux falls as $L/(4\pi d^2)$ for isotropic emission. A black-body star's peak wavelength decreases as temperature rises according to Wien's law.

9702-25-C1

ANSWER

Calibrate instruments, record raw data with uncertainties and consistent precision, plot a graph with labelled axes and appropriate scale, then calculate gradient using a large triangle. Repeat measurements to identify anomalies.

9702-26-C2

ANSWER

Plan to vary one independent variable while controlling others.

9702-26-C1

ANSWER

$T = 2.90 \times 10^{-3}/(500 \times 10^{-9}) = 5800$ K.

9702-25-P

ANSWER

State a testable relation, specify controls, propose apparatus and a safe repeatable method.

9702-27-C1

ANSWER

Longer total interval reduces reaction-time percentage error; control amplitude and bob/string setup or release method.

9702-26-P

ANSWER

A. String length if studying length
Why: Plan to vary one independent variable while controlling others.

9702-26-M

ANSWER

Which mechanism, condition or distinction matters in P5 Planning, analysis and evaluation?

9702-27-C2

QUESTION

To test $F = kv^2$, what should be plotted against v^2 ?

- A Force F
- B Time t
- C Mass m
- D Velocity v

9702-27-M

QUESTION

A model predicts $F = kv^2$. State graph axes to determine k and describe a check on experimental validity.

9702-27-P

QUESTION

Plot F against v^2 ; gradient gives k ; inspect straight-line fit, intercept and uncertainties/repeats.

9702-27-P

ANSWER

A. Force F
Why: State a testable relation, specify controls, propose apparatus and a safe repeatable method.

9702-27-M

ANSWER

Linearise a relationship to extract constants from gradient/intercept, propagate uncertainties and compare predicted values within those uncertainties.
Improvements should target the dominant error mechanism.

9702-27-C2

ANSWER